

Home Lab Project - Initial Plan

Overview

This project describes how to set up a two-node HomeLab using Proxmox VE on two Dell OptiPlex 7010 (Ultra Small Form Factor) systems.

The goal is to self-host simple services like synced note-taking, file sharing, media streaming, and development environments; all using Free and Open Source Software (FOSS).

Hardware

Device	CPU	RAM	Storage
Dell OptiPlex 7010 (USFF)	Intel Core i5 (3rd Gen)	8 GB	256 GB SSD
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Terminology Used

Term	Meaning
Proxmox VE (PVE)	A Debian-based hypervisor OS installed on each server. It manages VMs and containers and provides a single web interface to control all nodes in your setup.
Virtual Machine (VM)	A full simulation of a computer (CPU, RAM, storage, OS). Resource-heavy because it runs an entire operating system inside Proxmox.
Container (CT / LXC)	A lightweight environment that isolates applications but shares the host kernel. Ideal for running apps efficiently without the overhead of a full OS.
Hypervisor	Software that creates and manages virtual machines and containers—in this case Proxmox VE .
Node	A physical machine (e.g., each Dell OptiPlex) running Proxmox VE. Each one becomes part of your cluster.
Proxmox Cluster	A group of Proxmox nodes linked together. It provides one unified management interface and allows migration of VMs/containers between nodes.
Cluster / Node	(Merged into definitions above – see Node and Proxmox Cluster .)
Shared Storage	A storage system accessible by multiple nodes, allowing features like live migration of VMs/containers.
NAS (Network Attached Storage)	A central storage device on your network where files (photos, videos, backups, etc.) are stored and accessed by other devices.
Self-Hosted	Running and managing your own applications or services instead of relying on cloud platforms.
FOSS	Free and Open Source Software — can be used, modified, and distributed freely.

FOSS Stack

Service Request	FOSS Application	What It Does	How We'll Run It
Hypervisor	Proxmox VE	The base OS for both PCs. Manages all your other services.	Bare Metal (Main OS)
NAS / File Sharing	Samba (in a CT)	The standard protocol for file sharing. This makes your "NAS" visible as a network drive on Windows, macOS, and Linux.	Debian CT
Movie/Video Streaming	Jellyfin	A fantastic media server (like a free, self-hosted Netflix). It scans your movies/shows and streams them to your devices.	Debian CT
Song Streaming	Navidrome	A lightweight music server. It scans your music, organizes it, and streams to your phone or web browser (Subsonic-compatible).	Debian CT
Developer OS	Ubuntu Server	The full Ubuntu operating system you requested for development work.	VM (Virtual Machine)
Synced Note-taking	Joplin Server	The backend server for the Joplin note-taking app. This will sync your notes between your phone, laptop, etc.	Debian CT
Remote Access	Tailscale	You're already using it! It creates a secure virtual network (a VPN) so you can access your lab from anywhere, <i>without</i> router access.	On Proxmox Hosts & in CTs
Remote Access via Internet	Access services from anywhere	Tailscale + Traefik / Nginx	Safe HTTPS access
AI (Experimental)	Ollama	A tool to run open-source large language models (LLMs) locally on your own hardware.	Debian CT
File Conversion	Pandoc	A tool to convert between file formats	Developer OS

Additional Recommended Services

Service	Description	FOSS Option
Git Server	Host your code repositories	Gitea
Personal Wiki	Keep class notes & documentation	HedgeDoc
Code Wiki Generator	Create wikis & documentation for repos using AI	deepWiki
Monitoring Dashboard	Track performance and uptime	Grafana + Prometheus
Backup Service	Automate backups and snapshots	Proxmox Backup Server or Rsync
Ad-blocker	Network-wide ad and tracker blocker for all devices; extremely lightweight and ideal for CTs.	Pi-hole
Password Manager	Self-hosted, secure, lightweight password manager compatible with Bitwarden clients.	Vaultwarden
HomePage	A clean, minimal dashboard that provides quick access to all your self-hosted services.	HomePage
MacOS VM	Mostly just for fun. Never really experienced MacOS before	MacOS Installer
Additional File Conversion	Another file converter alternative hosted locally	ConvertX
Wiki Creator	Ability to create wikis from a homepage	DeepWiki

The Step-by-Step Implementation Timeline

Phase 1: The Foundation (The Cluster)

1. **Back Up Your Data** Before beginning, **back up all important files** from the Ubuntu machine. Both PCs will be wiped during installation.
2. **Install Proxmox VE (Node 1)**
 - Download the Proxmox VE ISO.
 - Flash it to a USB using Balena Etcher.
 - Install it on the first Dell PC. This establishes your **first hypervisor node**.
3. **Install Proxmox VE (Node 2)** Install Proxmox on the second PC the same way as Node 1. This prepares it for clustering.
4. **Create the Cluster**
 - On Node 1: Datacenter → Cluster → Create Cluster
 - On Node 2: Join the cluster using Node 1's join key. Your two machines now form a **Proxmox cluster** with unified control.
5. **Set Up Storage**
 - Use each node's 256 GB SSD as local storage.
 - On `pve-1`, create a main storage directory such as:

```
/mnt/storage
```

This becomes your media and shared data root.

Phase 2: Essential Services

6. **Remote Access (Tailscale)** Install Tailscale **directly on both Proxmox hosts**. This gives you secure, encrypted access to the cluster from anywhere.
7. **File Server (Samba CT)**
 - On `pve-1`, create a **Debian CT** with **512 MB RAM**.
 - Bind-mount `/mnt/storage` from the host into the container.
 - Install and configure Samba. You now have a simple NAS-like shared folder.
8. **Note Syncing (Joplin Server CT)**
 - On `pve-1`, create another **Debian CT** (512 MB RAM).
 - Install Docker (optional) and deploy Joplin Server. Your notes can now sync privately across devices.

Phase 3: The “Fun” Services

9. **Dashboard (Homepage CT)**
 - Create a small **Debian CT** (256 MB RAM).
 - Install Homepage. This becomes your unified dashboard for all services.
10. **Music Server (Navidrome CT)**
 - Create a **Debian CT** (512 MB RAM) on `pve-1`.
 - Bind-mount:

```
/mnt/storage/Music
```
 - Install Navidrome. This becomes your personal Spotify.
11. **Media Server (Jellyfin CT)**
 - Create a **Debian CT** (1 GB RAM).
 - Bind-mount your media directories, e.g.:

```
/mnt/storage/Movies
```
 - Install Jellyfin. This becomes your personal Netflix.

Phase 4: The “Work” Services

12. **Developer OS (Ubuntu VM)**
 - On `pve-2`, create an Ubuntu Server VM.
 - Recommended: **2 vCPUs, 2 GB RAM, 20 GB disk**.

- SSH into it for dev work. This gives you a proper Linux coding environment.

13. Ad-Blocking (Pi-hole CT)

- On pve-2 , create a **Debian CT** (512 MB RAM).
- Install Pi-hole. This provides whole-network ad and tracker blocking.

Phase 5: The Experiment

14. AI Experiment (Ollama CT)

- On pve-2 , create a **Debian CT** (≈2 GB RAM).
- Install Ollama.

Important: 3rd-gen i5 is not suited for modern AI workloads. Small models *will* run, but responses may take minutes. This is purely for experimentation and learning.

Potential Hardware Upgrades

1. **RAM:** The OptiPlex 7010 USFF has 2 slots and officially supports a maximum of 16GB (2x 8GB sticks) of **DDR3-1600** RAM. You can find used DDR3 RAM *very* cheaply on sites like eBay. Upgrading even one node to 16GB would make a *massive* difference.
2. **Storage:** 256GB is small for a media server. A 1TB or 2TB SATA SSD is a great upgrade. A cheaper alternative is a large USB 3.0 external hard drive. You can plug it into one Proxmox node, mount it, and use that as your main `/mnt/storage` pool.

Documentation Links

- Proxmox VE: [Installation Guide](#)
- Proxmox Cluster: [Cluster Manager Guide](#)
- Jellyfin: [Documentation](#)
- Navidrome: [Documentation](#)
- Joplin Server: [Installation Guide](#)
- Ollama: [GitHub/Documentation](#)
- Pi-hole: [Documentation](#)
- Vaultwarden: [Wiki/Guides](#)
- Tailscale: [Knowledge Base](#)

Starter Instructions

Goal: Install Proxmox VE on Node 1 and Create Your First VM

1. Download Proxmox VE ISO

- Visit <https://www.proxmox.com/en/downloads>
- Download the latest Proxmox VE Installer ISO.4

2. Create Bootable USB

- On any computer, use [Balena Etcher](#) to flash the ISO to a USB stick (≥2 GB).

3. Boot the OptiPlex from USB

- Insert USB → restart → press `F12` → choose `Boot from USB` .

4. Install Proxmox VE

- Select `Install Proxmox VE` .
- Choose your 256 GB SSD as install target.
- Create a root password and record it.
- Set hostname (e.g., `pve-node1.local`).
- Assign static IP (optional) or use DHCP.

5. Access the Web UI

- After reboot, note the IP (e.g., `192.168.x.x`).
- On your laptop, open browser → `https://<IP>:8006`
- Login:

```
Username: root
Password: (the one you set)
```

6. Create a VM

- Click "Create VM".
- Upload an Ubuntu Server ISO to Proxmox storage.
- Assign 2 vCPUs, 4 GB RAM, 20 GB disk.
- Boot the VM and install Ubuntu.

7. Test Networking

- Inside Ubuntu VM:

```
ping google.com
```

If it works, your networking bridge is correct.

8. Optional: Install Proxmox Updates

```
sudo apt update \&\& sudo apt full-upgrade -y
```